

1 Introduction

We study the interpolated stochastic heat and wave equation (ISHWE) for spatial dimension on $\mathbb{R}^d$,

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0 & b \in (0, 1] \\ u(0, \cdot) = u_0, \quad \partial_t u(0, \cdot) = v_0 & b \in (1, 2). \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

2 Preliminaries

For any $t \in [0, T]$ and $x \in \mathbb{R}^d$, we write (assuming $\beta \in (1, 2)$)

$$u(t, x) = J_0(t, x) + I(t, x)$$

where

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + \frac{\partial}{\partial t}[Z(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ I(t, x) &= \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy). \end{aligned}$$

1and2normYZ

Lemma 2.1 (Theorem 4.1 and Lemma 5.5 [1]). *Assume that $d < 2\alpha$, $\beta \in (0, 2)$, and $\gamma \geq 0$. Then*

$$\int_{\mathbb{R}^d} Y(t, x) dx = t^{\beta+\gamma-1}$$

and

$$\int_{\mathbb{R}^d} Y^2(t, x) dx = C_{\#} t^{2(\beta+\gamma-1)-d\beta/\alpha},$$

for all $t > 0$, where

$$C_{\#} := \frac{2}{\Gamma(d/2)(4\pi)^{d/2}} \int_0^\infty u^{d-1} E_{\beta, \beta+\gamma}^2(-u^\alpha) du.$$

Now further assume that $\beta \in (1, 2)$, then

$$\int_{\mathbb{R}} Z(t, x) = t^{\lceil \beta \rceil - 1} = t.$$

Remark 2.2. When $\alpha = \beta = 2$, $\gamma = 0$ and $d = 1$ then Lemma 2.1 gives us that $\int_{\mathbb{R}} G^2(t, x) dx = t/2$ which coincides with the $L^2(\mathbb{R})$ norm of the wave kernel.

In the proof of the next proposition, we will use the following two facts:

$$\sup_{t \geq 0} t^k e^{-at} = k^k (ea)^{-k} \text{ for } k \in \mathbb{N} \text{ and } \sup_{t \geq 0} \int_0^t s e^{-as} ds = a^{-2} \text{ for } a > 0. \quad (2.1) \quad \boxed{\text{E:SUPidenti}}$$

Proposition 2.3 (Proposition 3.2 [2]). *Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Then there exists a universal constant $K := K(\alpha, \beta, \gamma) > 0$ such that for any*

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-1)+\alpha(2\gamma-1))}.$$

and

$$p \in \left[2, \frac{a^{(\beta(2\alpha-1)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (2.2) \quad \boxed{\text{E:Nbdd}}$$

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (2.3) \quad \boxed{\text{E:Pnormtbdd}}$$

Proof. First we fix an $a > 0$ and $p \in [2, \infty)$. Using Lemma 2.1 we see that,

$$\begin{aligned} |J(t, x)| &= \begin{cases} |(Z(t, \cdot) * u_0)(x)| & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + \frac{\partial}{\partial t}(Z(t, \cdot) * u_0)(x)| & \beta \in (1, 2) \end{cases} \\ &\leq \begin{cases} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty & \beta \in (0, 1] \\ t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \end{aligned}$$

Now using (2.1), we see that

$$\begin{aligned} N_{a,p}(J) &\leq \begin{cases} \sup_{t \geq 0} \sup_{x \in \mathbb{R}} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty e^{-at} & \beta \in (0, 1] \\ \sup_{t \geq 0} \sup_{x \in \mathbb{R}} (t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty) e^{-at} & \beta \in (1, 2) \end{cases} \\ &= \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \quad (2.4) \quad \boxed{\text{E:Jbdd}}$$

Now we find an upper bound for $N_{a,p}(I)$. By simultaneously applying the Burkholder-Davies-Gundy inequality and triangle inequality, we get that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}} dy Y^2(t-s, x-y) \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^t ds \int_{\mathbb{R}} dy Y^2(t-s, x-y) \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying Lemma 2.1 and the Lipschitz property of σ we get that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_{\#} \int_0^t (t-s)^{2(\beta+\gamma-1)-\beta/\alpha} ds + 8pL^2(\sigma)N_{a,p}^2(u) \int_0^t ds \int_{\mathbb{R}} dy e^{2as} Y^2(t-s, x-y).$$

Now under Dalang's condition [1], $(1/\alpha - 2)\beta - 2\gamma < -1$, the first integral above can be calculated as

$$\int_0^t (t-s)^{2(\beta+\gamma-1)-\beta/\alpha} ds = \frac{\alpha t^{(2-1/\alpha)\beta+2\gamma-1}}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta}.$$

By applying Lemma 2.1, we can calculate the second integral as

$$\int_0^t ds \int_{\mathbb{R}} dy e^{2as} Y^2(t-s, x-y) = C_{\#} \int_0^t ds e^{2as} (t-s)^{2(\beta+\gamma-1)-\beta/\alpha}.$$

Putting this together gives us that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_{\#} \frac{\alpha t^{(2-1/\alpha)\beta+2\gamma-1}}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta} + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^t ds e^{2as} (t-s)^{2(\beta+\gamma-1)-\beta/\alpha}.$$

Now by multiplying both sides by e^{-2at} , we see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_{\#}\alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta} t^{(2-1/\alpha)\beta+2\gamma-1} e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^t ds e^{-2a(t-s)} (t-s)^{2(\beta+\gamma-1)-\beta/\alpha}. \end{aligned}$$

We now do a change of variables and again recall the definition of the Gamma function to see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_{\#}\alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta} t^{(2-1/\alpha)\beta+2\gamma-1} e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^\infty du e^{-2a(u)} (u)^{2(\beta+\gamma-1)-\beta/\alpha}. \\ &= \frac{8pc^2(\sigma)C_{\#}\alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta} t^{(2-1/\alpha)\beta+2\gamma-1} e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#}(2a)^{1+(1/\alpha-2)\beta-2\gamma}\Gamma[(2-1/\alpha)\beta-1+2\gamma] \end{aligned}$$

where the last line is true if Dalang's condition [1] holds. If we now apply (2.1) and also recall that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, then we see that

$$\begin{aligned} N_{a,p}(I) &\leq c(\sigma) \sqrt{\frac{8pC_{\#}\alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta}} [(2-1/\alpha)\beta/2 + \gamma - 1/2]^{(2-1/\alpha)\beta/2+\gamma-1/2} (ea)^{1/2-\gamma-(2-1/\alpha)\beta/2} \\ &\quad + L(\sigma)N_{a,p}(u) \sqrt{8pC_{\#}(2a)^{1+(1/\alpha-2)\beta-2\gamma}\Gamma[(2-1/\alpha)\beta-1+2\gamma]}. \end{aligned}$$

If we now let

$$\begin{aligned} K := \max \left\{ \sqrt{\frac{8C_{\#}\alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta}} \left(\frac{(2-1/\alpha)\beta/2 + \gamma - 1/2}{e} \right)^{(2-1/\alpha)\beta/2+\gamma-1/2}, \right. \\ \left. \left(\frac{1}{2} \right)^{(2-1/\alpha)\beta/2+\gamma-1/2} \sqrt{8C_{\#}\Gamma[(2-1/\alpha)\beta-1+2\gamma]} \right\} \end{aligned}$$

then,

$$N_{a,p}(I) \leq \frac{K\sqrt{p}}{a^{(2-1/\alpha)\beta/2+\gamma-1/2}} (c(\sigma) + L(\sigma)N_{a,p}(u)). \quad (2.5) \quad \boxed{\text{E:Ibdd}}$$

Now by combining (2.4) and (2.4), we see that

$$N_{a,p}(u) \leq \begin{cases} (\lceil\beta\rceil - 1)^{\lceil\beta\rceil-1} (ea)^{1-\lceil\beta\rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil\beta\rceil - 1)^{\lceil\beta\rceil-1} (ea)^{1-\lceil\beta\rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \quad (2.6)$$

$$+ K \frac{\sqrt{p}c(\sigma)}{a^{(2-1/\alpha)\beta/2+\gamma-1/2}} + K \frac{\sqrt{p}L(\sigma)}{a^{(2-1/\alpha)\beta/2+\gamma-1/2}} N_{a,p}(u). \quad (2.7) \quad \boxed{\text{E:Nbddgen}}$$

Now choose $a > 0$ such that

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-1)+\alpha(2\gamma-1))}.$$

This implies that the following interval is nonempty:

$$\left[2, \frac{a^{(\beta(2\alpha-1)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right].$$

In addition, for any p in this interval, we have that

$$\frac{\sqrt{p}L(\sigma)K}{a^{(2-1/\alpha)\beta/2+\gamma-1/2}} \leq \frac{1}{2},$$

and we also have

$$\frac{\sqrt{p}}{a^{(2-1/\alpha)\beta/2+\gamma-1/2}} \leq \frac{1}{2KL(\sigma)}.$$

This along with (2.7) implies that

$$N_{a,p}(u) \leq 2 \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} + \frac{c(\sigma)}{L(\sigma)}$$

and this is precisely (2.2). Note that (2.3) follows immediately from (2.2). $\square$

References

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